

KHAWLA SARHAN

Full-Stack Software Engineer | Enterprise & Scalable Systems

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PROFESSIONAL SUMMARY

Informatics Engineering graduate specializing in **Full-Stack Software Engineering & Scalable Web Architecture**. Hands-on experience architecting high-concurrency enterprise platforms (ERP), real-time applications, and complex financial systems using **Next.js 15, NestJS 11, PostgreSQL, Prisma ORM, and Docker**. Proven ability to translate complex business logic into production-ready software featuring transactional integrity, robust security (RBAC/JWT), and modern DevOps pipelines.

CORE TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript (ES6+), HTML5, CSS3
- **Frontend Development:** Next.js 15, React.js, Zustand, TanStack Query, Redux Toolkit, Tailwind CSS, Shadcn UI
- **Backend & Architecture:** NestJS 11, Node.js, Express.js, Prisma ORM, RESTful APIs, RBAC Security, Microservices Concepts
- **Databases & Real-Time:** PostgreSQL (Complex Aggregations & Transactions), MongoDB, Socket.IO (WebSockets)
- **DevOps & Infrastructure:** Docker, Nginx Reverse Proxy, CI/CD (GitHub Actions), Cloudflare, Cloudinary, Render

SELECTED PROJECTS

1. InvoSpark ERP — Enterprise Financial & Multi-Branch Inventory System (2026)

- **Technologies:** NestJS 11, Next.js 15, Prisma ORM, PostgreSQL, Socket.IO, Docker, TypeScript
- **Enterprise Financial Engine:** Architected full multi-branch accounting and inventory workflows with database transaction management and complex PostgreSQL aggregation queries via Prisma ORM.
- **Real-Time Synchronization:** Integrated Socket.IO for instant multi-terminal stock adjustments, live branch notifications, and audit logging.
- **Security & Infrastructure:** Implemented granular Role-Based Access Control (RBAC), JWT authentication flows, DTO validations, and containerized the architecture using Docker.

2. Taskly — Enterprise Task Management & Real-Time Ecosystem (2026)

- **Live Demo:** task-manager-frontend-Opvi.onrender.com
- **Real-Time Engine:** Architected a high-concurrency synchronization layer using Socket.IO, enabling instant task updates, live comments, and presence tracking across clients.
- **State & Cache Optimization:** Optimized frontend responsiveness using TanStack Query for intelligent server-state caching and Zustand for persistent session management.
- **DevOps Pipeline:** Fully containerized using Docker with Nginx reverse proxy and automated CI/CD deployment via GitHub Actions.

3. KH Media App — Real-Time Social Platform (2025)

- **Live Demo:** kh-media-app.pages.dev
- **Real-Time Core:** Implemented live chat, user presence, and instant alerts using Socket.IO with JWT-authorized WebSocket handshakes.
- **Authentication & Security:** Engineered token-based authentication (JWT) with automated Email OTP verification logic.
- **Performance:** Applied Next.js Server Components, Image Optimization, and lazy loading for high-speed content delivery.

4. Full-Stack E-commerce Platform (Client + Admin Portal) (2025)

- **Live Demo:** mernstack-e-commerce-store.pages.dev
- **Admin Management:** Implemented RBAC-protected administrative control over inventory, orders, and user data lifecycles.
- **Data Processing:** Engineered multi-faceted product filtering and search capabilities using MongoDB Aggregation Pipelines.
- **State & Checkout:** Integrated Redux Toolkit for client cart state management alongside secure payment processing workflows.

PROFESSIONAL EXPERIENCE

Full-Stack Software Engineer (Self-Directed Enterprise Projects) | 2023 - Present

Damascus, Syria

- **Enterprise Systems Development:** Designed and built end-to-end applications ranging from MERN real-time systems to enterprise financial ERP platforms.
- **Architecture & Clean Code:** Applied software engineering design patterns, modular architecture, and transactional database schemas.
- **DevOps Integration:** Containerized applications using Docker, managed environment configurations, and set up automated GitHub Actions pipelines.

IT Specialist | 2016 - 2023

Government Sector | Damascus, Syria

- Administered network infrastructure, systems availability, and technical support across enterprise units.
- Developed strong systematic debugging, network protocol analysis, and hardware/software integration skills.

EDUCATION

Bachelor's Degree in Informatics Engineering | University of Aleppo | 2010 - 2015

- Core Focus:** Software Engineering, Database Systems, Computer Networks, Data Structures & Algorithms.

ADDITIONAL QUALIFICATIONS

- Languages:** Arabic (Native), English (Professional Working Proficiency)
- Key Strengths:** Analytical Problem-Solving, System Architecture, Modern Web Protocols, Rapid Tech Adaptability.